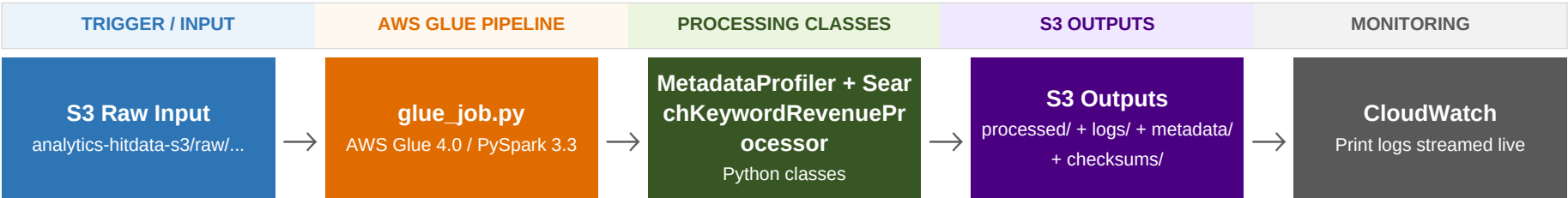


Architecture Overview

Search Keyword Revenue Pipeline — AWS Glue + S3



S3 artifacts written per execution:

Prefix	File Pattern	Written at Step	Purpose
processed/YYYY/MM/DD/	YYYY-MM-DD_SearchKeywordPerformance.tab	Step 4	Final ranked revenue output (tab-delimited)
logs/YYYY/MM/DD/	run-summary-YYYY-MM-DD-HH-MM-SS.json	Step 6	Run status, counts, output key
logs/YYYY/MM/DD/	pipeline-YYYY-MM-DD-HH-MM-SS.log	Step 7	Full pipeline log (mirrors CloudWatch)
metadata/YYYY/MM/DD/	metadata-log-YYYY-MM-DD-HH-MM-SS.json	Step 2	Column stats and schema mismatch details
checksums/	<sha256>.json	Step 5	Idempotency guard — maps file hash to output keys

Data Flow Detail

Step-by-step data transformation from raw hit-level file to ranked revenue output.

Step	Name	Input	Key Transformation	Output Artifact
0	Checksum Guard	s3://...raw/.../data.tab	Stream 8 MB chunks. Compute SHA-256. Check checksum against expected value (if found) and --force=false.	checksums/region/Extraction/.../data.tab
1	Read Source	S3 .tab file	spark.read.csv(sep=lt, quote="\", escape="\", multiLineCache=US568). Data Failure	Cached US568. Data Failure
2	Metadata Profiling	Cached DF + metadata2conf.json	distributed agg passes: min/max/null counts, then metadataDistC. De/MT/CD/metadata2conf.jsonfig.	metadataDistC. De/MT/CD/metadata2conf.jsonfig.
3	Business Transform	Cached DataFrame	Extract domain+keyword from referrer. Identify purchase event in sum revenue from product_list[3]. Last-touch Window join on IP.	reset(event=1) sum revenue from product_list[3]. Last-touch Window join on IP.
4	Write Output	result_df	coalesce(1). Write TSV to temp prefix. Copy part file processed.YYYY/MM/DD/prefix-YY-MM-DD_SearchKeywordPerformance.tab	processed.YYYY/MM/DD/prefix-YY-MM-DD_SearchKeywordPerformance.tab
5	Register Checksum	checksum + output/metadata2conf.json	add data object to checksums/<hash>.json with output_checksum/sha_key/first_processed_at_utc.	checksums/<hash>.json with output_checksum/sha_key/first_processed_at_utc.
6	Write Run Summary	Pipeline metrics	Build summary dict: status, records_processed, output_logs/YYYY/MM/DD/processed_at_utc.json	logs/YYYY/MM/DD/processed_at_utc.json
7	Flush Pipeline Log	PIPELINE_LOG_LINES	Join all log lines with newlines. s3.put_object as plain text	logs/YYYY/MM/DD/pipeline-<ts>.log

S3 Bucket Structure

analytics-hitdata-s3 — complete prefix layout and ownership.

Prefix	Written By	Read By	Contents
raw/YYYY/MM/DD/HH-MM-SS/	External / upstream process	glue_job.py Step 1	Source Adobe Analytics hit-level .tab files. Input only — never modified by the pipeline
processed/YYYY/MM/DD/	glue_job.py Step 4	End consumers / analysts	Final ranked output. One tab file per calendar day per source file.
processed/YYYY/MM/DD/_tmp_output_	glue_job.py Step 4	glue_job.py Step 4	Temporary Spark part files. Created and deleted within the same run.
logs/YYYY/MM/DD/	glue_job.py Steps 6–7	Operators / on-call team	run-summary-<ts>.json and pipeline-<ts>.log. One pair per execution. Never overwritten
metadata/YYYY/MM/DD/	glue_job.py Step 2	Data quality team	metadata-log-<ts>.json. Column stats, schema mismatch details, row count.
checksums/	glue_job.py Step 5	glue_job.py Step 0	<sha256>.json idempotency registry. Maps file hash to output key and first_processed
config/	Manual / deployment script	glue_job.py Step 2	metadata_config.json passed via --extra-files. Defines expected schema.
scripts/	Manual / deployment script	AWS Glue runtime	glue_job.py — the Glue job script itself.